WEEK -2 |WEEKLY REPORT

Network Traffic Analysis, Nmap and Windows Logging

Reporting period: Week 2 • Submission deadline: 30 October 2026, 11:59 PM IST

Prepared for: DG Interns Hub | Evidence-based lab summary

Executive Summary

Week 2 focused on network packet analysis, Nmap scanning, Windows Security and System events, and introductory PowerShell threat hunting. The submission contains theory notes, six TryHackMe room writeups and completion screenshots, a Wireshark practical writeup, lab captures, scan outputs, event evidence, and two reference cheat sheets.

Learning Objectives

Understand TCP handshakes and flags; recognise normal and suspicious DNS, HTTP, HTTPS/TLS, ARP, ICMP and SMB traffic; use Wireshark capture and display filters; distinguish Nmap scan types; understand Windows processes, services, registry, accounts and logs; identify relevant Security Event IDs, logon types, and SOC baselines.

Lab Environment

The practical lab used Kali Linux (192.168.56.102), Ubuntu (192.168.56.101) and Windows 11 (192.168.56.104) on a VirtualBox host-only network. NAT connectivity was used where necessary. Apache was installed on Ubuntu for HTTP traffic generation. All scans and log generation described here were carried out against owned lab virtual machines.

Theory and TryHackMe

Eight separate theory notes cover the assigned subjects. Six TryHackMe rooms were documented: Nmap, Live Host Discovery, Basic Port Scans, Windows Basics, Windows Fundamentals 1, and Windows Fundamentals 2. Completion screenshots are stored alongside the individual writeups. The room writeups discuss concepts; commands from the separate local lab are not presented as proof of particular TryHackMe steps.

Wireshark Traffic Capture

From Kali, ICMP echo traffic, HTTP requests to the Ubuntu Apache server, and DNS lookups were generated and captured. Screenshots show TCP SYN / SYN-ACK / ACK, HTTP GET, DNS query, and ICMP echo. Evidence includes DNS_Traffic.pcapng and ICMP_HTTP_TCP_Traffic.pcapng. These captures illustrate normal traffic patterns that can later be contrasted with suspicious activity.

Nmap Scanning and Packet Evidence

Five scan categories were recorded against both Ubuntu and Windows: host discovery (-sn), TCP SYN (-sS), service/version (-sV), operating system detection (-O), and aggressive detection (-A). Each scan has TXT (-oN) and XML (-oX) output: 20 scan-output files total. Ubuntu SYN scan identified TCP/80 open (HTTP) and 999 closed TCP ports. The Windows SYN scan used -Pn and reported all 1,000 tested TCP ports filtered (no response), consistent with firewall filtering; this does not establish that Windows has no services. A separate PCAPNG and screenshots show SYN/RST-related scan traffic.

Target	Observed SYN scan result	Evidence
Ubuntu 192.168.56.101	80/tcp open (http); 999 closed	Ubuntu_SYN_Scan.txt
Windows 192.168.56.104	1000 filtered TCP ports	Windows_SYN_Scan.txt

Windows Event Viewer

Screenshots document Event ID 4625 (failed logon), 4624 (successful logon), 4720 (user created), 4732 (user added to Administrators), 7045 (service installed, System log), and 1102 (Security log cleared). A separate screenshot shows logon type 5 (service logon) for the observed 4624 event; it should not be described as an interactive user login. A Security log EVTX backup was saved before the log was cleared. The lab user was subsequently removed from Administrators, and the dummy service was deleted.

PowerShell Hunting

Get-WinEvent with -FilterHashtable was used to read Event ID 4625 from the saved EVTX file after the active Security log had been cleared. Three failed-logon events were found and exported to Failed_Logins_4625.csv. Group-Object showed three failed logons for account kesheth. These events were generated in the controlled lab and do not independently indicate an intrusion.

Account	Failed logons (4625)
kesheth	3

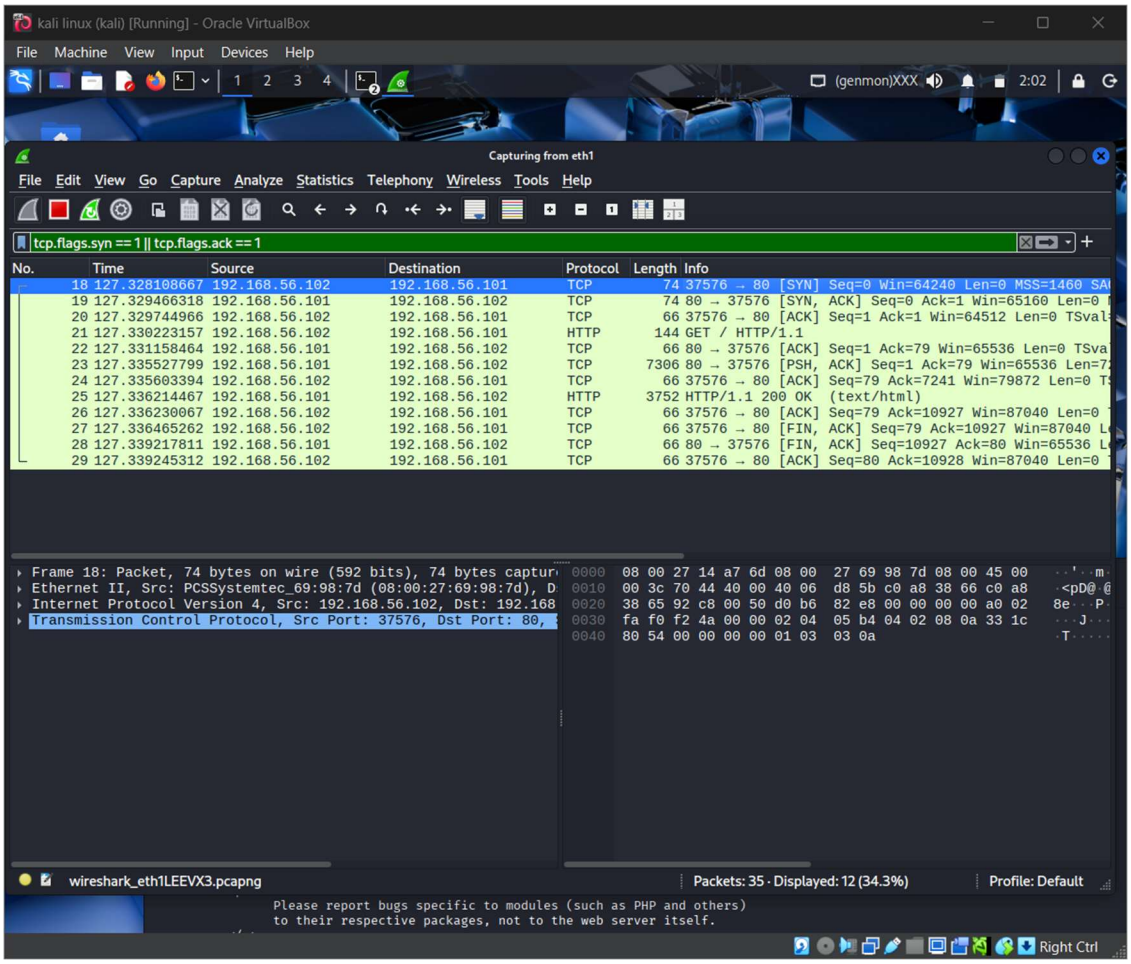
Analyst Findings and Limitations

The Ubuntu web service was reachable on TCP/80; the Windows SYN scan produced filtered-port results. Packet captures illustrate handshake, HTTP, DNS and ICMP patterns. Security logs show controlled account and service changes plus three failed logons. Screenshots and scan files provide reproducible evidence, but this report does not claim a confirmed compromise. LinkedIn publication and Google Drive submission cannot be verified from the provided archive.

Deliverables and Conclusion

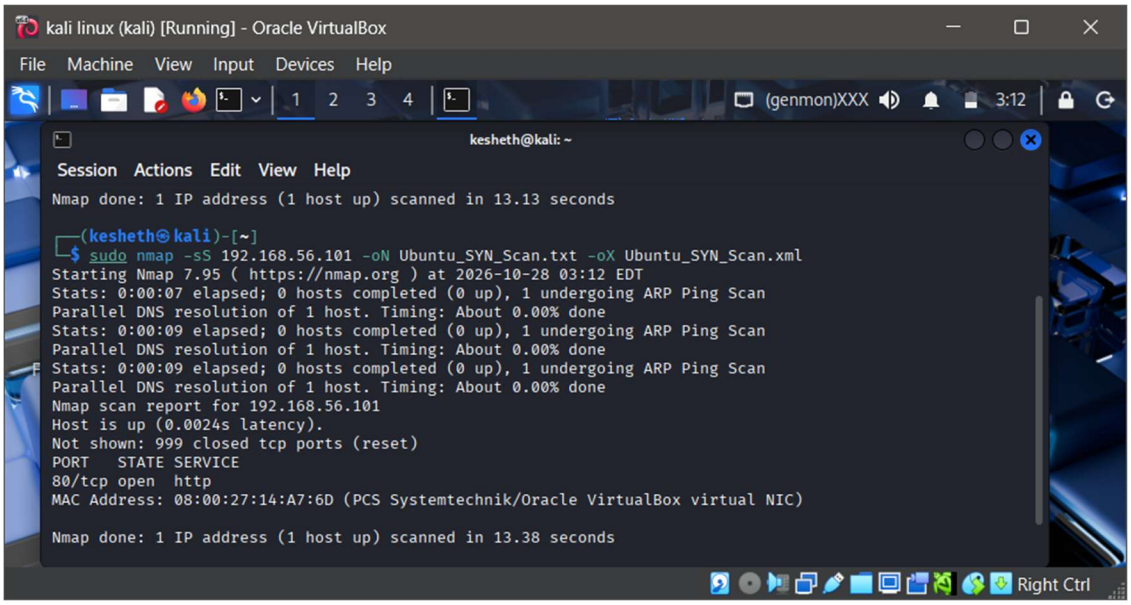
Submission contents: 8 theory notes; 6 TryHackMe writeups with screenshots; 1 Wireshark writeup; packet captures and scan outputs; Nmap/Wireshark/Windows/PowerShell screenshots; Security EVTX backup and failed-logon CSV; 2 cheat sheets; this weekly report in DOCX and PDF; a PowerPoint presentation; and updated README. Week 2 established foundational visibility into network and Windows endpoint telemetry ahead of Week 3 IDS and endpoint work.

Selected Evidence — TCP handshake



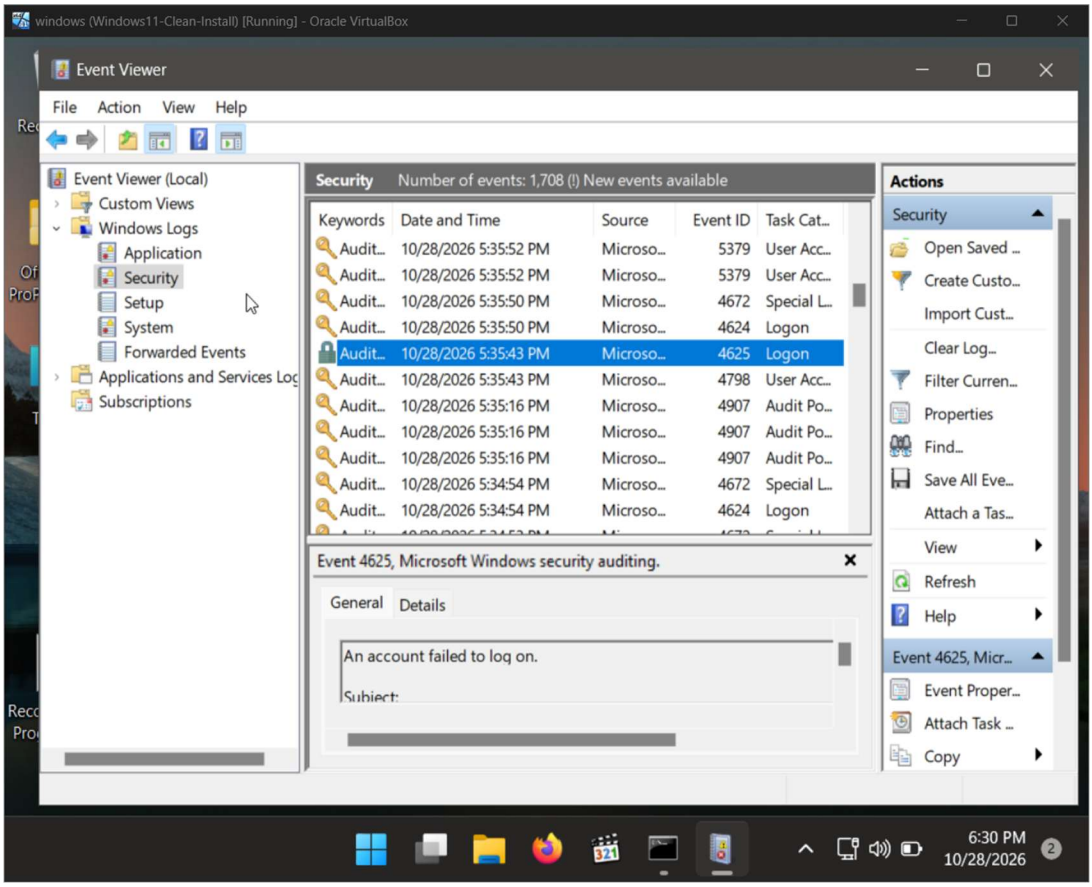
Source: Findings/Wireshark/TCP_Handshake.png

Selected Evidence — Ubuntu Nmap SYN scan



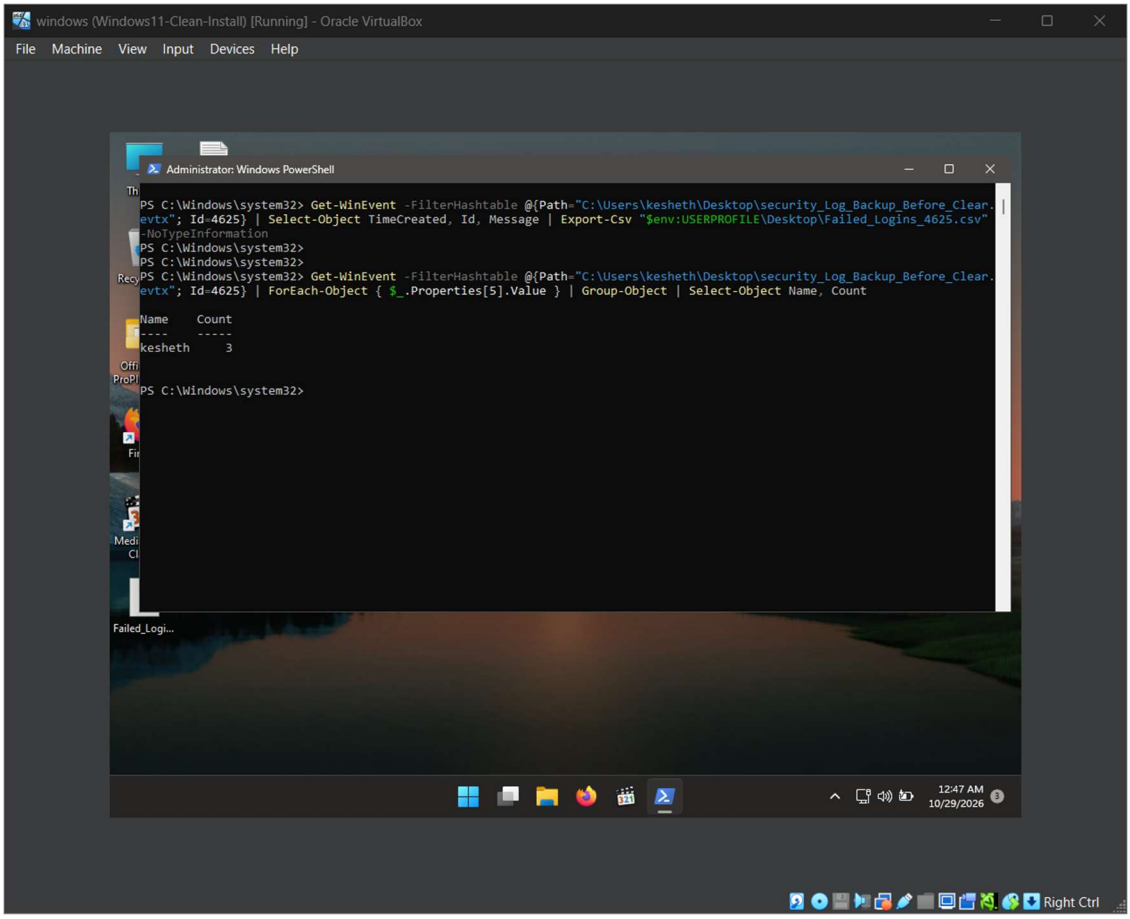
Source: Findings/Nmap/Ubuntu/02_Ubuntu_SYN_Scan.png

Selected Evidence — Event Viewer 4625



Source: Findings/Windows Events Viewer/01_Failed_Logon_Event_4625.png

Selected Evidence — PowerShell grouped failures



Source: Findings/PowerShell/02_PowerShell_Failed_Logins_Per_Account.png

Evidence note: The source archive contains the underlying screenshots, PCAPNG, scan outputs and log files. All observations in this report are scoped to the controlled virtual lab.